Parametric Equations

The points on a graph can be defined in terms of a third variable, say t.

So x and y are expressed in terms of t over an interval of t-values :

$$x = f(t) \text{ and}$$
$$y = g(t) \; ; \; a \leq t \leq b$$

The third variable is called the parameter.

Given the parametric equations:

$$x = 3t + 1 \qquad 0 \le t \le 4$$
$$y = t + 5$$

Describe the graph and write a Cartesian equation for the graph.

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Ex: $x = t + 1$ and $y = t^2 + 1$

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Given $x = 1 - t$ and $y = \sqrt{t}$, $t \geq 0$

Write y as a function of x:

$$x = 2t - 4 \quad and \quad y = 3 + t^2$$

What type of graph do the parametric equations:

$x = \cos t$ and $y = \sin t$ represent? $0 \le t \le \pi$

Describe the graph and write a Cartesian equation for the graph.

$x = \sin(t)$ and $y = \cos(t)$

Given a curve defined parametrically in terms of $x(t)$ and $y(t)$ we have

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

Find dy/dx given $x(t) = t^2-4t$ and $y(t) = 5 + 12t – t^3$

Given the parametric equations:

$$x = 3t + 1 \qquad 0 \le t \le 4$$
$$y = t + 5$$

Write an equation of the line tangent to the curve described above at t = 1.

Write an equation of the line tangent to $x = 4e^{2t}$ and $y = 3t^4$ at $t = \ln 3$